

Section 5: Nmap & Lua (NSE Scripts)

Task 5.1: Basic Nmap Scanning

Target used for all scans: **scanme.nmap.org**

1. Basic Ping Scan Command

Used:

```
nmap -sn scanme.nmap.org
```

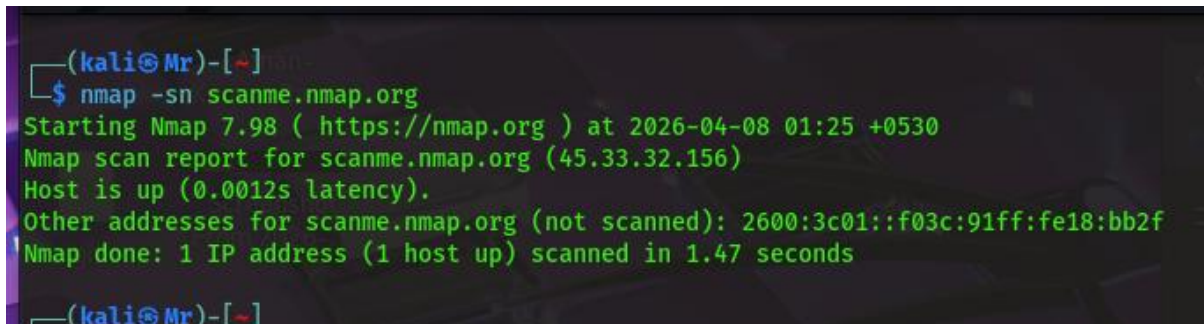
Explanation:

This command performs a ping scan to check whether the target host is alive or reachable. It does not scan ports but only determines host availability.

Expected Result:

The output shows whether the host is up along with its IP address.

Screenshot:



```
(kali@Mr)-[~]  
$ nmap -sn scanme.nmap.org  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-08 01:25 +0530  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.0012s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Nmap done: 1 IP address (1 host up) scanned in 1.47 seconds  
  
(kali@Mr)-[~]
```

2. TCP SYN Scan (Top 100 Ports)

Command Used:

```
nmap -sS scanme.nmap.org
```

Explanation:

This command performs a TCP SYN scan (also called half-open scan). It scans the top 100 most common ports and identifies whether they are open, closed, or filtered. It is faster and stealthier than full TCP scans.

Expected Result:

The output lists open ports such as 80 (HTTP) and 22 (SSH), along with their states.

Screenshot:

```
(kaliⓂMr)-[~]
$ nmap -sS scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-08 01:27 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0016s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 511 filtered tcp ports (no-response), 485 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite
Nmap done: 1 IP address (1 host up) scanned in 383.05 seconds
```

3. Service Version Detection on Port 80

Command Used: `nmap -p 80 -sV`

scanme.nmap.org

Explanation:

This scan targets port 80 specifically and attempts to detect the service and its version running on that port.

Expected Result:

The output shows details such as the web server type (e.g., Apache, nginx) and version.

Screenshot:

```
(kaliⓂMr)-[~]
$ nmap -p 80 -sV scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-08 01:36 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.00057s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE VERSION
80/tcp    filtered http

Service detection performed. Please report any incorrect results at https://nmap.org
Nmap done: 1 IP address (1 host up) scanned in 1.42 seconds
(kaliⓂMr)-[~]
```

4. OS Detection Scan Command

Used:

```
nmap -O scanme.nmap.org
```

Explanation:

This command attempts to identify the operating system of the target by analyzing network responses and TCP/IP fingerprinting.

Expected Result:

The output shows the guessed operating system and accuracy percentage.

Screenshot:

```
(kali@kali)~$ nmap -O scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-08 01:36 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.011s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 769 filtered tcp ports (no-response), 227 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp   open  nping-echo
31337/tcp  open  Elite
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (90%), Oracle Virtualbox (88%), Slirp (88%), QEMU (86%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (90%), Oracle Virtualbox Slirp NAT bridge (88%), QEMU user mode network gateway (86%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 472.13 seconds
```

5. Scan Using Default NSE Scripts

Command Used:

```
nmap -sC scanme.nmap.org
```

Explanation:

This scan uses default Nmap Scripting Engine (NSE) scripts to gather additional information about the target, such as vulnerabilities, services, and configurations.

Expected Result:

The output includes script results like HTTP titles, SSL information, and other useful details.